

Práctica 3: Formulario

- I. $\frac{d}{dx}(c) = 0.$
- II. $\frac{d}{dx}(x) = 1.$
- III. $\frac{d}{dx}(u + v - w) = \frac{du}{dx} + \frac{dv}{dx} - \frac{dw}{dx}.$
- IV. $\frac{d}{dx}(cu) = c \cdot \frac{du}{dx}.$
- V. $\frac{d}{dx}(uv) = u \cdot \frac{dv}{dx} + v \cdot \frac{du}{dx}.$
- VI. $\frac{d}{dx}(u^n) = nu^{n-1} \frac{du}{dx}.$
- VII. $\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \cdot \frac{du}{dx} - u \cdot \frac{dv}{dx}}{v^2}.$
- VIII. $\frac{d}{dx} \left(\frac{u}{c} \right) = \frac{1}{c} \cdot \frac{du}{dx}.$
- IX. $\frac{dy}{dx} = \frac{dy}{dv} \cdot \frac{dv}{dx}.$
- X. $\frac{d}{dx}(y) = \frac{1}{\frac{dx}{dy}}.$
- XI. $\frac{d}{dx}(\ln v) = \frac{1}{v} \cdot \frac{dv}{dx}.$
- XII. $\frac{d}{dx}(\log v) = \frac{\log e}{v} \cdot \frac{dv}{dx}.$
- XIII. $\frac{d}{dx}(a^v) = a^v \ln a \cdot \frac{dv}{dx}.$
- XIV. $\frac{d}{dx}(e^v) = e^v \cdot \frac{dv}{dx}.$
- XV. $\frac{d}{dx}(u^v) = vu^{v-1} \frac{du}{dx} + \ln u \cdot u^v \frac{dv}{dx}.$
- XVI. $\frac{d}{dx}(\sin v) = \cos v \cdot \frac{dv}{dx}.$
- XVII. $\frac{d}{dx}(\cos v) = -\sin v \cdot \frac{dv}{dx}.$
- XVIII. $\frac{d}{dx}(\tan v) = \sec^2 v \cdot \frac{dv}{dx}.$
- XIX. $\frac{d}{dx}(\cot v) = -\csc^2 v \cdot \frac{dv}{dx}.$
- XX. $\frac{d}{dx}(\sec v) = \sec v \tan v \cdot \frac{dv}{dx}.$
- XXI. $\frac{d}{dx}(\csc v) = -\csc v \cot v \cdot \frac{dv}{dx}.$
- XXII. $\frac{d}{dx}(\arcsin v) = \frac{1}{\sqrt{1-v^2}} \frac{dv}{dx}.$
- XXIII. $\frac{d}{dx}(\arccos v) = \frac{-1}{\sqrt{1-v^2}} \frac{dv}{dx}.$
- XXIV. $\frac{d}{dx}(\arctan v) = \frac{1}{1+v^2} \frac{dv}{dx}.$